

School of Computer Science and Engineering

MCA , 3rd Sem

**Internship Project Report
Data Analytics Intern**

**Organization Name-
Codec Technologies Pvt. Ltd.**

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Introduction

- **Overview:** Summarizes the industrial data analytics internship conducted at Codec Technologies Pvt. Ltd.
- **Core Objective:** Bridges raw operational data with executive-level decision-making through standard analytical pipelines.
- **Dual-Tool Architecture:** Combines open-source Python libraries for data processing with Microsoft Power BI for dynamic dashboarding.
- **Application:** Applied to enterprise sales and operational datasets to identify trends, performance anomalies, and optimization areas.

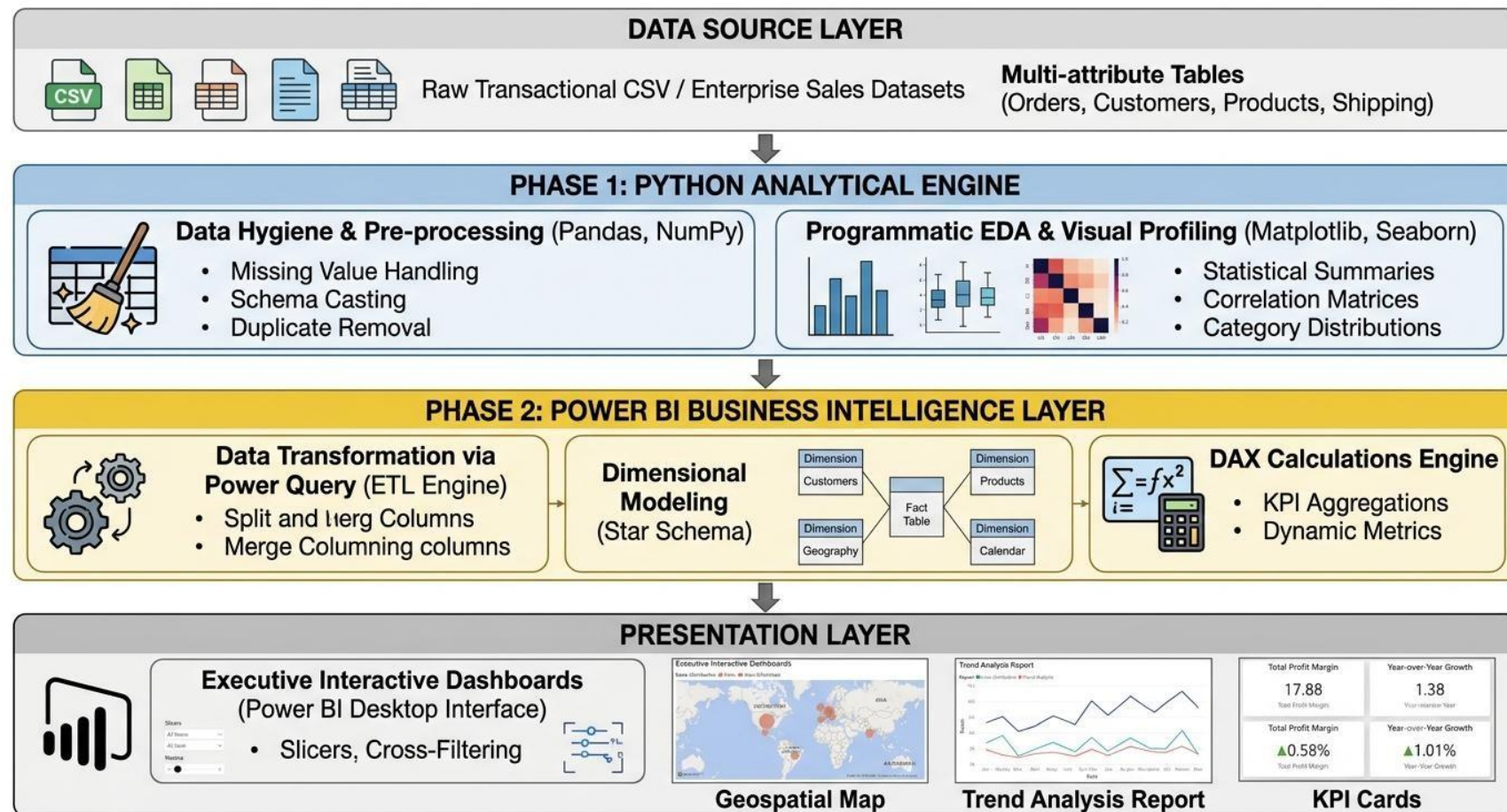
Problem Statement

- **Data Quality Bottlenecks:** Raw enterprise datasets frequently contain null values, duplicate orders, and inconsistent schema formatting.
- **Static Reporting Limits:** Spreadsheet-based traditional reporting lacks interactive filtering, cross-highlighting, and real-time drill-down capabilities.
- **Operational Visibility Gap:** Stakeholders struggle to track regional profit losses, delivery lag times, and customer segment metrics effectively.
- **Target Need:** An automated ETL pipeline to clean data programmatically and output interactive, executive-grade dashboards.

Scope of the Project

- **In-Scope Elements:**
 - Ingestion and validation of multi-attribute enterprise transactional records.
 - Automated Python routines for missing value imputation, duplicate purging, and schema casting.
 - Power Query ETL, Star Schema data modeling, and custom DAX metric calculations.
 - Multi-page interactive dashboards featuring dynamic slicers, drill-downs, and geospatial maps.
- **Out-of-Scope Elements:**
 - Real-time live database streaming connections (analysis performed on static enterprise exports).
 - Predictive machine learning algorithm deployment (focus remains strictly on Descriptive and Diagnostic analytics).

System Architecture



* A Modular end-to-end Analytical Pipeline (Python + Power BI) - Architecture Diagram

Technologies and Tools Used

- **Programming Language:** Python (Executed in Jupyter Notebook / Anaconda environment).
- **Data Processing Libraries:** Pandas (Dataframe manipulation) and NumPy (Numerical operations).
- **Data Visualization Libraries:** Matplotlib (Static plots) and Seaborn (Statistical distributions).
- **Business Intelligence Platform:** Microsoft Power BI Desktop.
- **Analytics Engines:** Power Query (ETL) and Data Analysis Expressions (DAX).

Methodology

- **Phase 1 — Programmatic Pre-processing (Python):**
 - Ingestion: Loading flat files into Pandas dataframes for baseline checks.
 - Hygiene: Automated null value handling, casting date/numeric types, and purging duplicate records.
 - EDA: Executing univariate, bivariate, and heatmap visualizations to analyze distributions and correlations.
- **Phase 2 — Business Intelligence (Power BI):**
 - ETL: Power Query transformations, attribute column splitting, and table merging.
 - Data Modeling: Constructing a Star Schema (central Fact table linked to Customer, Product, Geography, and Date dimension tables).
 - Calculations & UI: Writing custom DAX measures for KPIs and designing dynamic, multi-page visual reporting interfaces.

References

- McKinney, W. (2017). Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython (2nd ed.). O'Reilly Media.
- Ferrari, A., & Russo, M. (2016). Analyzing Data with Microsoft Power BI and Power Pivot for Excel. Microsoft Press.
- Russo, M., & Ferrari, A. (2019). The Definitive Guide to DAX: Business Intelligence with Microsoft Power BI, SQL Server Analysis Services, and Excel (2nd ed.). Microsoft Press.
- Harris, C. R., et al. (2020). Array programming with NumPy. Nature, 585(7825), 357–362.
- Microsoft Corporation (2024). Power BI Desktop Data Modeling & DAX Technical Documentation.
- Codec Technologies Pvt. Ltd. (2026). Data Analytics Industrial Internship Curriculum & Standards.

Thank You